

File 65: Commercial Angling Micro-SaaS & Trout Unlimited Scientific Alliance Playbook

Commercial and Scientific Alliance Standard: This document outlines the monetization models and non-profit collaboration strategy for Blue Ridge Stream Restoration's real-time water quality telemetry network. By connecting local, off-grid LoRa-mesh and cellular stream monitors to commercially viable Micro-SaaS widgets, Hardware-as-a-Service (HaaS) leases, and corporate ESG offset mechanisms, we turn environmental data into a sustainable revenue generator. This playbook details our target commercial vectors, lists direct scientific and technical contacts within Trout Unlimited (TU) at both national and regional levels, and maps out an advanced system engineering roadmap to elevate our hydrological infrastructure to USGS-grade precision.

I. Commercial Applications & Angler Monetization Channels

Fluvial data-sharing is not only a tool for environmental stewardship; it represents a highly scalable business opportunity. Managing Director Hunter Morris can monetize our streamside sensor network through three distinct commercial channels:

1. Fly Shop & Guide Service Micro-SaaS (Embedded Web Widgets)

- **The Problem:** Local fly shops (e.g., Unicoi Outfitters in Helen, Cohutta Fishing Company in Blue Ridge) and drift-boat guide services spend hours compiling daily "fishing reports" to drive website traffic. These reports are often subjective and lack real-time water quality metrics.
- **The SaaS Solution:** Blue Ridge licenses dynamic, interactive, and beautifully styled web widgets that fly shops embed directly on their homepages or blogs.
- **Data Delivered:** Real-time stream temperature, gauge stage level (wadeability), and turbidity indexes powered by our local streamside sensors.
- **Monetization Model:** A monthly subscription fee of **\$29.00 to \$49.00 USD per widget** per stream chapter.
- **Guide Alert Network:** An add-on premium **SMS text alert broadcast service (\$9.99/month)** for local guides, sending instant push notifications when a stream drops to a safe wading level or triggers a "Hoot Owl" heat restriction.

2. Private Lodge & Trout Estate Hardware-as-a-Service (HaaS)

- **The Problem:** High-end private trout clubs and luxury fly-fishing estates (such as those along the Soque River or the Upper Toccoa) stock thousands of dollars of trophy rainbow and brown trout. Stream temperature spikes or sudden storm spates (floods) can decimate their fish populations or cause severe bank erosion.
- **The HaaS Solution:** Blue Ridge leases fully assembled off-the-shelf WisBlock streamside NEMA nodes under a **Hardware-as-a-Service (HaaS) lease**.
- **Services Provided:** Blue Ridge field crews install and maintain the off-grid nodes, calibrate the hydrostatic stage level and optical dissolved oxygen sensors monthly, and provide the estate manager with a private

dashboard.

- *Monetization Model:* **\$149.00 to \$249.00 USD per month** per property.
- *Actionable Alerts:* Automatic notifications to shut down automatic feeders when temperatures exceed 68°F (trout cannot digest food efficiently in warm water) or warn of incoming high-flow events that require deploying debris deflectors.

3. Exurban Data Center ESG & Thermal Credit Offsets

- **The Problem:** Hyperscale data center operators (QTS, Microsoft, Google) consume millions of gallons of local groundwater daily for evaporative cooling, lowering aquifer water tables and decreasing coldwater stream baseflows. This leads to regulatory scrutiny and public pushback.
- **The B2B ESG Solution:** Blue Ridge stream sensors act as the third-party validation network for a proprietary **Thermal Stream Offset program**.
- *Data Delivered:* Continuous thermal attenuation verification, proving that our geomorphic canopy restorations are cooling streams by an amount equivalent to the data center's hydrologic footprint.
- *Monetization Model:* High-ticket annual consulting contracts and recurring "Thermal Offset Credit" verification fees (**\$15,000 to \$50,000+ USD per year**) billed to the corporate sustainability departments of tech conglomerates.

II. Trout Unlimited Scientific & Technical Alliance Directory

To secure fast-track approvals, validate our geomorphic designs, and gain warm introductions to large mountain landowners, Hunter Morris must align Blue Ridge's data network with the technical and scientific leaders of **Trout Unlimited (TU)**.

1. National Scientific & GIS Leadership Contacts

These individuals manage TU's national conservation databases, GIS mapping layers, and environmental monitoring standards:

- **Warren Colyer — National Restoration Director**
 - *Sourcing Role:* Leads national stream habitat restoration projects.
 - *Collaboration Pitch:* Present our geomorphic woody-debris and Rhododendron stabilization standards as a blueprint for Western-style wood placements adapted to Southern Appalachian high-gradient channels.
 - *Email:* warren.colyer@tu.org
- **Daniel Dauwalter — Fisheries Science Director**
 - *Sourcing Role:* Coordinates scientific protocols for TU's fisheries research and community science databases.
 - *Collaboration Pitch:* Share our JSON REST API stream telemetry payloads directly with TU's national water quality registries to study long-term thermal refuge durability.
 - *Email:* daniel.dauwalter@tu.org
- **Louis Jochems — GIScience Director**
 - *Sourcing Role:* Oversees TU's spatial analysis, GIS platforms, and stream conservation mapping suites.
 - *Collaboration Pitch:* Deliver real-time GeoJSON station endpoints mapping North Georgia coldwater headwaters to integrate directly into TU's national aquatic conservation maps.

- Email: louis.jochems@tu.org
- **Matthew Mayfield — Senior GIS Analyst**
- *Sourcing Role:* Specialized spatial analysis mapping trout habitat vulnerabilities.
- *Collaboration Pitch:* Partner to cross-reference our 100 North Georgia large river tracts GIS sourcing data with TU's regional habitat vulnerability overlays.
- Email: matthew.mayfield@tu.org

2. Regional Southern Appalachian Sourcing & Project Leaders

These staff members manage on-the-ground restoration funding, regulatory permissions, and forest service partnerships in our direct geographic footprint:

- **Jacob Rash — Program Director, Southern Appalachians**
- *Sourcing Role:* Directs all TU staff, funding, and public agency partnerships across the Southern Appalachian brook trout range.
- *Collaboration Pitch:* Propose a joint "Georgia Spawning Bio-Reserve Initiative," leveraging Warnell graduate interns to install off-the-shelf WisBlock sensors at key restoration reaches.
- Email: jacob.rash@tu.org
- **Jeff Wright — Southern Appalachian Project Manager**
- *Sourcing Role:* The primary field coordinator managing stream habitat restoration, culvert replacements, and barrier removals in North Georgia and Western North Carolina.
- *Collaboration Pitch:* Coordinate our instream excavation schedules with TU's regional culvert replacement schedules to maximize downstream sediment capture.
- Email: jeffrey.wright@tu.org

3. Georgia Volunteer Council Conservation Contacts

These grassroots leaders manage local chapter budgets, coordinate volunteer cleanups, and maintain deep relationships with local North Georgia mountain landowners:

- **Emily Owenby — Conservation Chair, Georgia Council of TU**
- *Sourcing Role:* Coordinates volunteer restoration workdays and state conservation grants.
- *Collaboration Pitch:* Invite local chapters to participate in spring stocking and bio-assessment surveys on our restored Anderson Creek reach.
- Email: conservation@georgiату.org
- **Rodney Tumlin — Chairman, Georgia Council of TU**
- *Sourcing Role:* Leads the 12 active local chapters across the state of Georgia.
- *Collaboration Pitch:* Establish Blue Ridge Stream Restoration as a sponsor of the Georgia Council's annual conservation fundraisers.
- Email: chairman@georgiату.org
- **Steve Westmoreland — Georgia National Leadership Council (NLC) Representative**
- *Sourcing Role:* Connects Georgia volunteer efforts with national policy directors.
- *Collaboration Pitch:* Advocate for standardizing digital streamside telemetry tools at national NLC summits.
- Email: national_leadership@georgiату.org

III. System-Level Engineering Upgrades: Taking Infrastructure to the Next Level

To transition our current off-grid telemetry network from a functional prototype to a world-class, commercial-grade hydrological platform, we must implement these four advanced engineering upgrades:

1. Locally Trained LSTM Machine Learning Thermal Predictive Models

- **The Upgrade:** Instead of simply reacting when water temperatures cross critical trout stress thresholds, we train a localized **Long Short-Term Memory (LSTM) recurrent neural network**.
- **How it Works:** The model processes historical 15-minute stream temperatures, ambient air temperatures (from local NOAA feeds), upstream baseflow stage, and LiDAR-derived canopy shade indexes.
- **The Result:** Hunter Morris can predict "Hoot Owl" temperature violations **48 hours in advance**, allowing fly guides to proactively adjust client bookings and data centers to optimize their cooling operations ahead of thermal heat domes.

2. Rating Curve Volumetric Discharge (Q) Computations

- **The Upgrade:** Currently, our hydrostatic level sensors measure gauge stage height (H in meters). To match USGS standards, we must calculate volumetric discharge (Q in cubic feet per second - CFS) in real time.
- **How it Works:** Interns establish localized **stage-discharge rating curves** by taking manual flow-meter cross-section measurements during various flow levels. We fit the data to the standard power-law equation:

$$Q = C(H - a)^b$$

- **Where:** Q is discharge, H is the measured stage height, a is the stage of zero flow, and C and b are fitted coefficients unique to each geomorphic channel cross-section.
- **The Result:** The AWS Lambda function automatically converts the raw hydrostatic voltage into live CFS discharge figures, displaying USGS-grade flow data on our public B2B dashboard.

3. WisBlock Low-Power Deep Sleep Hardware Interrupts

- **The Upgrade:** To achieve absolute battery independence and support 5+ year operations on small 3.7V batteries, we implement hardware-based low-power interrupt routines.
- **How it Works:** Instead of keeping the MAX485 and boost converter rails connected to active GPIO pins, we route the WisBlock core's primary UART RX line directly to an external hardware interrupt pin (GPIO pin 9). The MCU remains in a **15 micro-amp deep sleep state**.
- **The Result:** The sensor node only wakes up when the RAK4631 internal RTC timer fires every 15 minutes, executes the 5-second sensor scan, broadcasts the encrypted LoRa hex payload, and instantly shuts down the entire voltage rail. This reduces total daily battery consumption by **98%**, allowing operation even under total canopy cover with zero solar input.

4. TimescaleDB Compression and Downsampling Policies

- **The Upgrade:** As our sensor network expands to 50+ streams, storing raw 15-minute readings will rapidly bloat database storage and slow down dashboard loading times.
- **How it Works:** We implement automated **TimescaleDB compression and downsampling policies** directly on the RDS PostgreSQL database:
- **Compression:** Automatically compress timeseries chunks older than 14 days using Timescale's columnar compression algorithms, reducing disk footprint by up to **90%**.

- *Downsampling*: Run a daily background worker that downsamples data older than 90 days into hourly averages, and data older than 365 days into daily averages, preserving historical baseline trends while keeping the database exceptionally lean and fast.

Developed by Blue Ridge Stream Restoration Technical Sourcing Operations. Pushed to remote origin under main.